Women Safety analytics & Threat detecttion

Divy Jetani - 202302626010030

Harsh Italiya - 202302626010027

Dhyey Chavadiya - 202302626010013

Project Guid: Dr. Aditi Joshi

Faculty of engineering [GLS University]

Introduction

- Women’s safety is a major concern in urban and semi-urban environments.
- Many incidents of harassment and violence happen in: Public places (streets, bus stops, markets)
- Semi-private environments (campuses, offices, hostels)
- Existing systems: CCTV surveillance – mostly passive, manually monitored. Women safety apps – mostly reactive, require user to press SOS.
- There is a need for a proactive, integrated system that: Detects threats automatically from CCTV.
- Connects to a mobile app for alerts and SOS.

Aim

"To design and develop an AI-powered women safety and threat detection system that analyzes CCTV footage in real time to detect potentially violent or threatening situations, and integrates with a mobile application to send rapid alerts to women, guardians and security authorities, focusing on campus-scale deployment."

The Aim is

- To build an integrated AI-based surveillance and alert system that:
  - Automatically detects violence or abnormal behavior from CCTV video.
  - Sends timely alerts through a mobile app and a security dashboard.
  - Enhances proactive safety and response time for women in campus environments.

Objective

1. Detect violent or threatening activities from CCTV using deep learning.
2. Generate risk scores and real-time alerts to security staff and women users.
3. Provide a mobile app with SOS button, live location sharing and alert notifications.
4. Build a web dashboard for monitoring incidents and managing alerts.
5. Analyze system performance (accuracy, latency, false positives).
6. Ensure basic privacy, security, and ethical compliance in the prototype.

Functional Requirements

- User registration and login (women users, admin/security).
- Mobile app features:
  - One-tap SOS.
  - Location sharing with server and guardians.
  - View recent incident alerts and risk zones.
- CCTV AI module:
  - Connect to CCTV / RTSP streams.
  - Process video frames and classify violence vs non-violence.
  - Trigger alerts with severity scores.
- Admin dashboard:
  - View, filter and manage alerts.
  - Play associated video clips/snapshots.
  - Change alert status (New → In Review → Escalated → Resolved).

Non-functional Requirements

- Performance:
  - Near real-time processing (alert within ~5-10 seconds).
- Accuracy:
  - At least ~80–85% accuracy on test datasets.
- Reliability:
  - System should not crash during continuous video streaming.
- Scalability:
  - Support multiple camera streams in future.
- Security & Privacy:
  - Password hashing, JWT authentication.
  - Access control for admin vs normal users

Resources

Tech Stack: Mobile: Flutter (Dart), Web: React.js, tailwind CSS, bootstrap, Backend: python, javascript, node.js, express.js, Database: postgreSQL, SQLite

Libraries: openCV, TensorFlow, scikit-learn, CNNs, YOLO, MobileNet, Socket IO

Services and APIs: Firebase authentication, firebase notifications, gemini API, API ninjas

Tools/Softwares: VS code, Android studio, Postman, Figma

Literature Review

Sr.	Title	Year	Methodology	Limitation
1	AI Tools and Applications for Women's Safety	2024	Survey, reporting analysis, AI-based campus safety	Limited to campus settings; user perception bias
2	A Study on AI-Powered Women's Safety App	2025	Mobile app development, location tracking, voice AI	Small dataset, hardware dependency
3	Creating a Safe Environment for Women: A Location-based Review Platform	2024	Crowdsourcing, location data, safety index calculation	Data reliability, real-time validation
4	A systematic review of ethical challenges and opportunities in AI for women’s safety	2023	Literature review, qualitative analysis	Lack of standardized ethical frameworks
5	A systematic review of ethical challenges and opportunities in AI for women’s safety	2022	Policy review, recommendations for ethical AI	General ethics, not application-specific

Literature Review

Sr.	Title	Year	Methodology	Limitation
6	Violence Detection in Surveillance Footage Using Deep Learning	2022	Deep learning (CNNs) for video classification	Dataset imbalance, false positives
7	Real-Time Women Safety System Using IoT and AI	2023	IoT devices, AI-based pattern detection	Device cost, maintenance, data security
8	Smart Wearable Device for Women Security Leveraging AI	2021	IoT wearable prototype, sensor fusion, ML inference	Wearability, power consumption, acceptance
9	Enhancing Women Safety via Predictive Analytics in Smart Cities	2024	Data mining, predictive modeling, city crime statistics	City-specific, data quality, predictive limits
10	Community-based Alert Systems Using Crowdsourcing and ML	2023	App-based alert system, ML on crowdsourced events	Information lag, crowdsourcing quality

Literature review (AI for Violence Detection)

- Research shows deep learning models can detect:
  - Fights, assaults, abnormal crowd behavior.
- Common datasets:
  - RWF-2000 – real-world fight videos from CCTV-like views.
  - UCF-Crime – large anomaly detection dataset.
- Crowd Violence.
- Common approaches:
  - 2D CNN + LSTM (spatial + temporal).
- Results: high accuracy (>80%) on benchmark datasets.

Literature review (Woman safety apps)

- Apps like 112 India, Safetipin, Noonlight, Kavalan SOS, Life360:
  - Provide SOS buttons.
  - Share location to contacts / control rooms.
  - Some use safety scoring for areas.
- Limitations:
  - Require user to manually trigger SOS.
  - Not connected to CCTV video analytics.
  - More reactive than proactive.

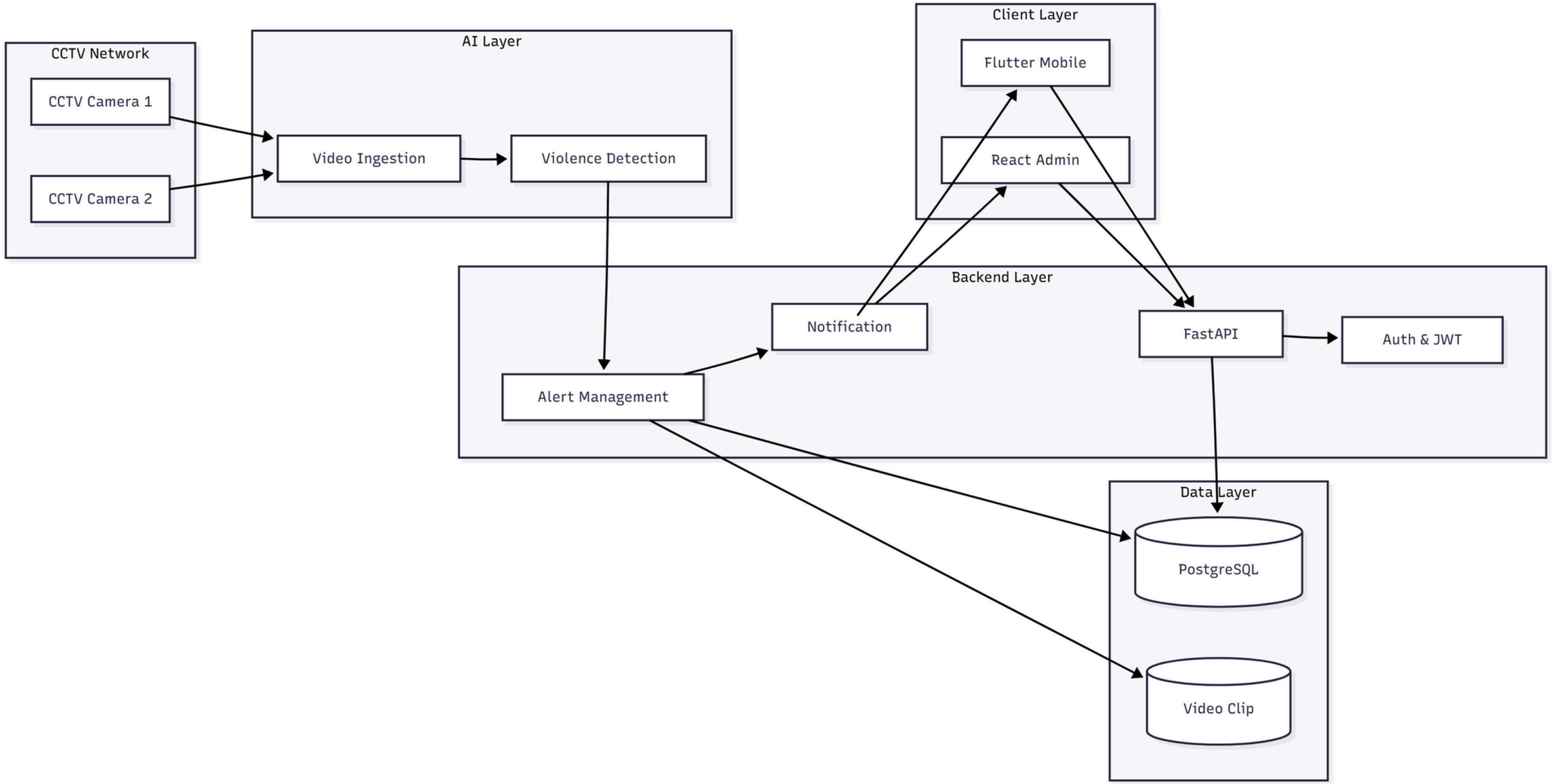
Comparisons

Criteria	Existing Women Safety Apps	CCTV Surveillance	Proposed System
Detection mode	Manual SOS	Human operators	AI + manual SOS
Proactive threat detection	No	Limited (depends on operator)	Yes (AI-based violence detection)
Integration with CCTV	No	Yes	Yes, AI-enabled
User-facing mobile app	Yes	No	Yes
Response time	Depends on user action + manual review	Depends on operator speed	Automated detection + instant alerts
Privacy considerations	Location data	Video surveillance	Both, with designed safeguards

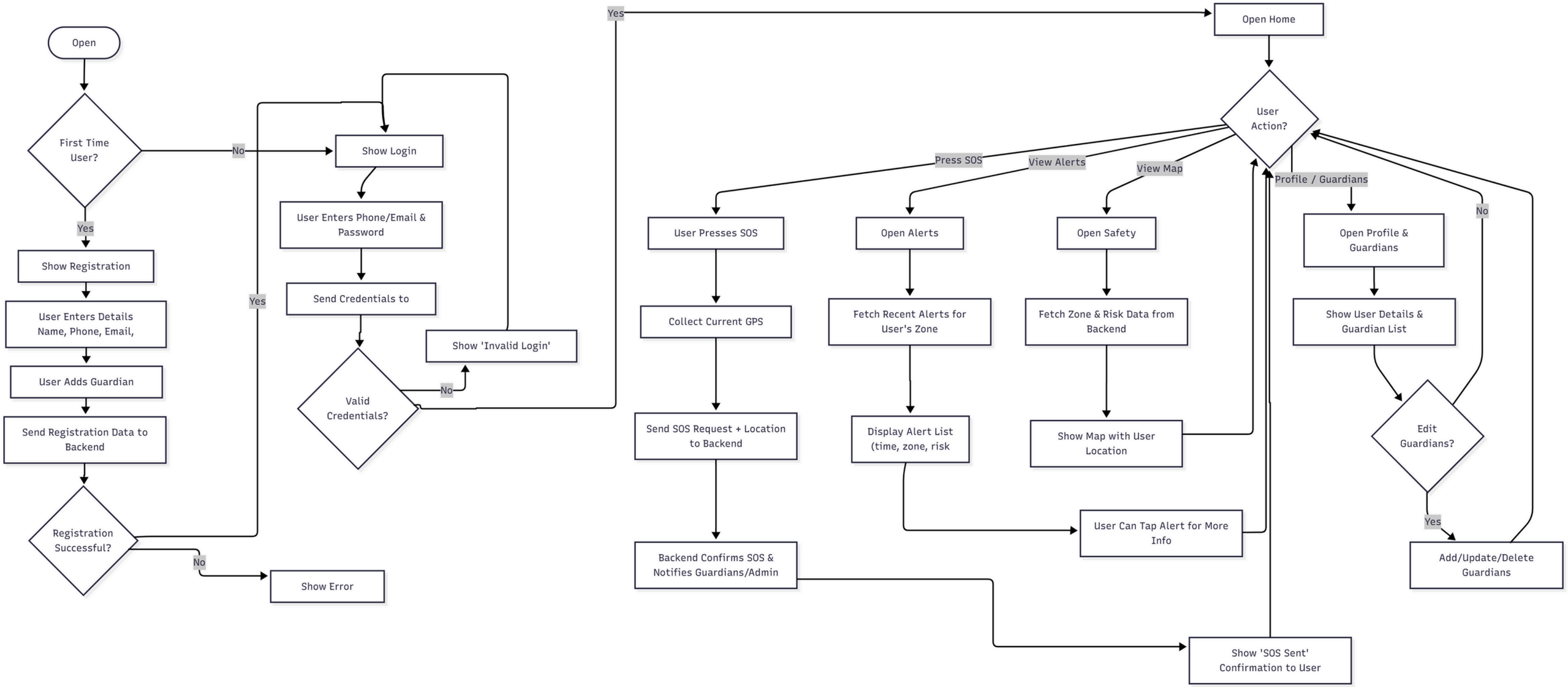
Proposed Solution

- AI-powered CCTV analysis:
  - Violence detection using 3D CNN / CNN+LSTM.
- Integrated alert engine:
  - Risk scoring based on AI confidence, zone, time.
  - Automatic alert generation.
- Mobile app for women users:
  - SOS, location sharing, alert notifications, risk view by zone.
- Web dashboard for security/admin:
  - Real-time alerts list and video clips.
  - Incident management and statistics.
- Focus on campus-scale deployment:
  - Universities, IT parks, large institutions.

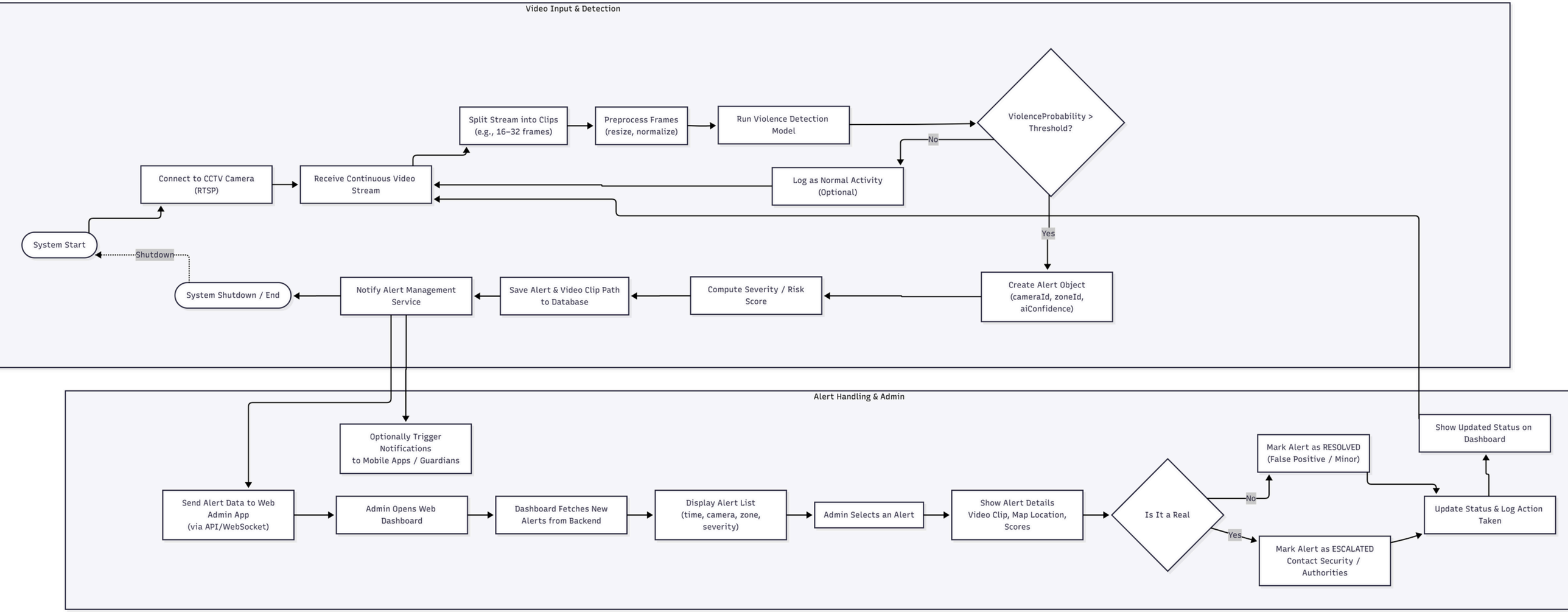
System Architechture



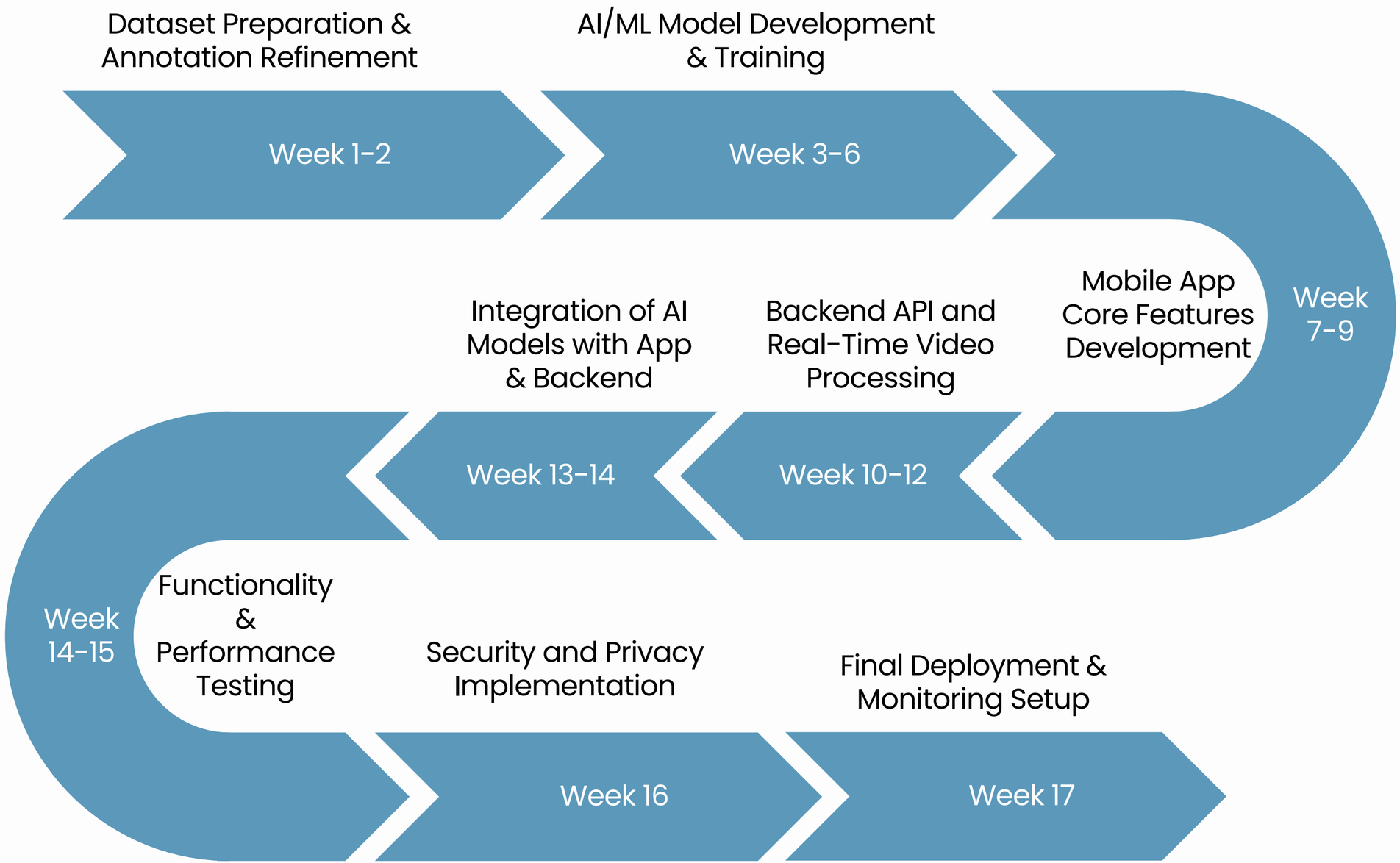
Flow chart for Mobile app



Flowchart for AI surveillance



Timeline


Dataset Preparation & Annotation Refinement
Week 1-2
AI/ML Model Development & Training
Week 3-6
Mobile App Core Features Development
Week 7-9
Backend API and Real-Time Video Processing
Week 10-12
Integration of AI Models with App & Backend
Week 13-14
Functionality & Performance Testing
Week 14-15
Security and Privacy Implementation
Week 16
Final Deployment & Monitoring Setup
Week 17

Conclusion

- System demonstrates how:
  - AI-based video analysis can add proactive intelligence to existing CCTV infrastructure.
  - Integration with a mobile app increases personal safety for women users.
- The system is:
  - Scalable to larger environments (e.g., city safe-city projects).
  - Extensible to include more features such as:
    - Gesture-based distress detection.
    - Integration with emergency police helplines.
- Future work:
  - Improve model generalization.
  - More robust privacy mechanisms.
  - Real-world pilot deployments with institutional partners.

References

1. [Negre et al.] Deep Learning for Violence Detection in Video, Pattern Analysis and Applications, 2020.
2. [Ullah et al.] Vision-based Violence Detection in Video – Survey, AI Review, 2021.
3. RWF-2000: An Open Large Scale Video Database for Violence Detection, 2020.
4. UCF-Crime Dataset: Real-World Anomaly Detection in Surveillance Videos.
5. Vidhyaa J.V. et al., Violence Detection in Videos using Conv2D VGG-19, 2021.
6. AURA – An AI-driven Mobile Safety Application for Women, 2025.
7. Official sites/articles for 112 India, Safetipin, Noonlight, Kavalan SOS, etc.
8. Articles on AI-powered Safe City CCTV deployments (e.g., Lucknow Safe City).

References

- AI Tools and Applications for Women's Safety, NGMC, 2024_
https://www.ngmc.org/cdn/uploads/iqac/C3/3_4_4/2023-2024/B_Com(BI)/2.pdf

- A Study on AI-Powered Women's Safety App, Scribd, 2025
https://www.scribd.com/document/866267811/A-Study-on-AI-Powered-Women-s-Safety

- Creating a Safe Environment for Women: Location-based Review Platform, SSGMCE, 2024
https://www.ssgmce.ac.in/uploads/UG_Projects/cse/202324/Project%20Report%20Gr.%20No.%2016_2023-24.pdf

- Systematic Review of Ethical Challenges and Opportunities in AI for Women's Safety, ScienceDirect, 2023
https://unesdoc.unesco.org/ark:/48223/pf0000391719

“Empowering women’s safety is empowering our future.”

Thank You